145470

Course Name: **MACHINE LEARNING**

Course Outcome

CO1- Understand the basic concepts of machine learning.
CO2-Apply the concepts of regression, classification, and re-sampling methods.
CO3-Design supervised and re-enforcement learning based solution.
CO4-Apply the ensemble methods for improving classification.
CO5-Identify the ways of feature extraction, reduction and selection.

Printed Pages: 02 **University Roll No. .....**

Mid Term Examination, Even Semester, 2025-26
B.TECH.(CS/Biotech), III Year, VI Semester
BCSE 0105 MACHINE LEARNING

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
Attempt all the questions section-wise

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	The following data shows the relationship between the number of hours studied (X) and marks obtained (Y): X: 2, 4, 6, 8 Y: 3, 5, Null, 10 First, impute the missing Y value using **forward fill** and then find the regression line Y=A.X+B	3	2	A	P
2	Explain how PCA helps in reducing dimensionality and improving model performance.	3	5	U	C
3	A logistic regression model is defined as: $P = \frac{1}{1 + e^{-(2+0.5X)}}$ a) Calculate the probability when $X = 2$. b) Determine whether the output class is **0 or 1** if the decision threshold is **0.5**.	3	2	A	P
4	Why is outlier detection important in machine learning? Name any two methods used for outlier detection.	3	1	U	C
5	An image-processing startup is developing a **classification system** that categorizes objects based on two extracted features X_1 and X_2. Each object belongs to one of two classes: **Red** or **Blue**. The existing labeled dataset is given below. A new object **P (5, 5)** is detected by the system. Identify the 3 nearest neighbors	3	3	A	P

Data Point	X_1	X_2	Class
A	2	4	Red
B	4	6	Blue
C	4	4	Red
D	6	8	Blue
E	8	6	Blue

and classify the new data point **P** using KNN Algorithm.

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Given actual values $Y = [2,4,6]$ and predicted values $\hat{Y} = [3,5,5]$. Calculate: a) Mean Absolute Error (MAE) b) Mean Squared Error (MSE)	5	1	E	P
7	a) How does classification differ from regression? b) What is the bias–variance trade-off? Explain its significance in model performance.	5	2	U	C
8	A university is developing a **machine learning model** to analyze students' preferences across different departments for academic planning. The dataset contains a **categorical feature** called *Department*, representing each student's preferred department. The dataset is shown below: a) What are the different methods of handling the categorical feature *Department*. b) Apply **one-hot encoding** to convert the categorical feature into numerical form and write the output.	5	1	A	P

Student	Department
S1	CSE
S2	ECE
S3	ME
S4	CSE
S5	ECE